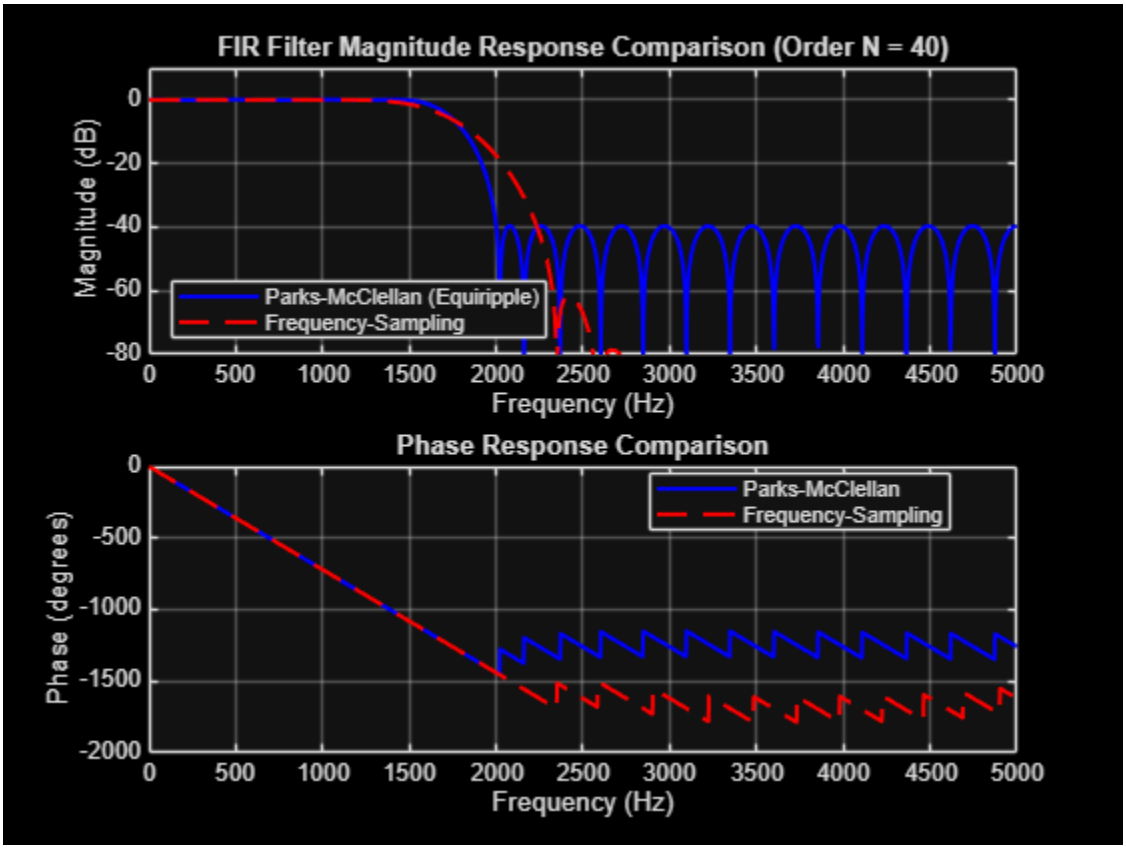
```
clear;
clc;
close all;

Fs = 10000;         % Sampling frequency (Hz)
fp = 1500;          % Passband edge frequency (Hz)
fs = 2000;          % Stopband edge frequency (Hz)
N = 40;             % Filter order

% Normalized frequency band edges (0 to 1)
f_bands = [0 fp fs Fs/2] / (Fs/2);

% Desired amplitudes at the band edges
a_bands = [1 1 0 0];

% ========================================
% 1. PARKS-MCCLELLAN (EQUIRIPPLE) DESIGN
% ========================================
b_pm = firpm(N, f_bands, a_bands);

% ========================================
% 2. FREQUENCY-SAMPLING DESIGN
% ========================================
b_fs = fir2(N, f_bands, a_bands);

% ========================================
% DISPLAY NUMERICAL OUTPUTS
% ========================================
fprintf('=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY-SAMPLING ===\n\n');

fprintf('Filter Order (N): %d\n', N);
fprintf('Number of Coefficients: %d\n', N+1);
fprintf('Sampling Frequency: %d Hz\n', Fs);
fprintf('Passband Edge: %d Hz\n', fp);
fprintf('Stopband Edge: %d Hz\n\n', fs);

fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---\n');
disp(b_pm(1:10));

fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---\n');
disp(b_fs(1:10));

% ========================================
% FREQUENCY RESPONSE CALCULATION
% ========================================
[H_pm, f] = freqz(b_pm, 1, 2048, Fs);
H_fs = freqz(b_fs, 1, 2048, Fs);

% Magnitude response in dB
H_pm_dB = 20*log10(max(abs(H_pm), 1e-10));
H_fs_dB = 20*log10(max(abs(H_fs), 1e-10));
```

```
% ========================================
% PLOTTING
% ========================================
figure('Name', 'Parks-McClellan vs Frequency-Sampling', ...
    'NumberTitle', 'off');

% Magnitude Response
subplot(2,1,1);

plot(f, H_pm_dB, 'b', 'LineWidth', 1.5);
hold on;
plot(f, H_fs_dB, 'r--', 'LineWidth', 1.5);

grid on;
legend('Parks-McClellan (Equiripple)', ...
    'Frequency-Sampling', ...
    'Location', 'southwest');

title(['FIR Filter Magnitude Response Comparison (Order N = ', ...
    num2str(N), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([0 Fs/2]);
ylim([-80 10]);

% Phase Response
subplot(2,1,2);

plot(f, unwrap(angle(H_pm))*180/pi, ...
    'b', 'LineWidth', 1.5);
hold on;
plot(f, unwrap(angle(H_fs))*180/pi, ...
    'r--', 'LineWidth', 1.5);

grid on;
legend('Parks-McClellan', 'Frequency-Sampling', ...
    'Location', 'best');

title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);

fprintf('\nFilter design and plotting completed successfully.\n');
```

```
=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY-SAMPLING ===

Filter Order (N): 40
Number of Coefficients: 41
Sampling Frequency: 10000 Hz
Passband Edge: 1500 Hz
Stopband Edge: 2000 Hz

--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---
```

```
  Columns 1 through 7

   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037

  Columns 8 through 10

    0.0142    0.0100   -0.0090

--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---
  Columns 1 through 7

    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008

  Columns 8 through 10

    0.0035    0.0029   -0.0039

Filter design and plotting completed successfully.
```



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